

translation of the form

$$\psi(z) = z + m\omega_1 + n\omega_2, \quad z \in \mathbb{C},$$

is a covering transformation. Let φ be an arbitrary covering transformation. Since $\varphi(0) \in \pi^{-1}(\{0\}) = L$, there are integers m and n such that $\varphi(0) = m\omega_1 + n\omega_2$. Then the covering transformation $\psi^{-1} \circ \varphi$ satisfies $(\psi^{-1} \circ \varphi)(0) = 0$. By the theorem, $\psi^{-1} \circ \varphi$ is the identity, and consequently, $\varphi = \psi$. Thus the covering transformations are precisely the translations by the lattice points in L .

Example. Let $\varphi : \mathbb{C}^* \rightarrow \mathbb{C}^*$ be a covering transformation for some covering map. Then φ is a fractional linear transformation, so φ has a fixed point. From the theorem we conclude that φ is the identity.

It turns out that every Riemann surface can be obtained from a simply connected surface by identifying points. We state the relevant theorem without proof, though the proof will be outlined in the exercises.

Theorem. *For each Riemann surface R , there is a covering map $\pi : S \rightarrow R$ of a simply connected Riemann surface S onto R . If $p_0 \in R$ and $q_0, q_1 \in \pi^{-1}(\{p_0\})$, there is a unique covering transformation φ of S such that $\varphi(q_0) = q_1$.*

The surface S is called the **universal covering surface** of R . The surface R can be regarded as a quotient surface of S , obtained by identifying two points q_0 and q_1 of S if there is a covering transformation $\varphi : S \rightarrow S$ such that $\varphi(q_0) = q_1$.

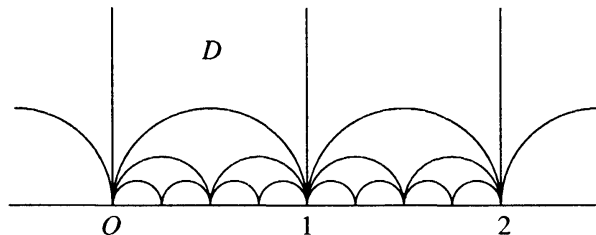
According to the uniformization theorem, there are only three possibilities for the universal covering surface: the open unit disk $\mathbb{D}$, the complex plane $\mathbb{C}$, and the Riemann sphere $\mathbb{C}^*$. Most surfaces have the unit disk as their universal covering surface.

Theorem. *The only Riemann surface having the Riemann sphere $\mathbb{C}^*$ as its universal covering surface is the sphere itself. The only Riemann surfaces having the complex plane $\mathbb{C}$ as universal covering surface are the complex plane itself, the punctured complex plane $\mathbb{C} \setminus \{0\}$, and tori. All other Riemann surfaces have the open unit disk as universal covering surface.*

The first statement is a consequence of the fact that the only covering transformation $\varphi : \mathbb{C}^* \rightarrow \mathbb{C}^*$ is the identity. For the second statement, one shows that the only analytic functions $\varphi : \mathbb{C} \rightarrow \mathbb{C}$ that can be covering transformations are translations.

Example. Consider the domain consisting of the points z in the upper half-plane $\mathbb{H}$ such that $0 < \operatorname{Re} z < 1$ and $|z - \frac{1}{2}| > \frac{1}{2}$. The domain D is a geodesic triangle with respect to the hyperbolic metric of $\mathbb{H}$, with vertices

at 0, 1, and $i\infty$. We reflect D in the circular boundary arc $|z - \frac{1}{2}| = \frac{1}{2}$, and we obtain a geodesic triangle with vertices at 0, $\frac{1}{2}$, and 1, and with sides the semicircles determined by $|z - \frac{1}{2}| = \frac{1}{2}$, $|z - \frac{1}{4}| = \frac{1}{4}$, and $|z - \frac{3}{4}| = \frac{1}{4}$. We continue reflecting in the semicircular sides. The reflected images of D are disjoint geodesic triangles that, together with their sides, fill out the semistrip $\{0 < \operatorname{Re} z < 1, \operatorname{Im} z > 0\}$. If we reflect D also in its vertical sides, and continue reflecting, we fill out the entire upper half-plane with the various reflections of D and its sides. The result is a “tiling” of the upper half-plane by geodesic triangles. Consider the Riemann map π of D onto the upper half-plane, normalized by $\pi(0) = 0$, $\pi(1) = 1$, and $\pi(i\infty) = i\infty$. By the Schwarz reflection principle, the map π reflects across the circle $|z - \frac{1}{2}| = \frac{1}{2}$ and maps the reflected geodesic triangle onto the lower half-plane, so that the semicircle where $|z - \frac{1}{4}| = \frac{1}{4}$ is mapped to $(-\infty, 0)$ and $|z - \frac{3}{4}| = \frac{1}{4}$ to $(1, +\infty)$. We continue reflecting, and we obtain an analytic function π on the upper half-plane that maps each reflection of D onto either the upper or the lower half-plane, and that maps a common side between two such reflections to one of the three intervals $(-\infty, 0)$, $(0, 1)$, and $(1, +\infty)$. If D_0 is an open disk disjoint from the real axis, then $\pi^{-1}(D_0)$ is a disjoint unit of domains, one in “every other” of the reflected images of D , each mapped one-to-one onto D_0 . Something similar happens for disks in $\mathbb{C} \setminus \{0, 1\}$ meeting $\mathbb{R}$. Thus π is a covering map of the upper half-plane onto the twice-punctured plane $\mathbb{C} \setminus \{0, 1\}$. It represents the upper half-plane as the universal covering surface of $\mathbb{C} \setminus \{0, 1\}$. The composition of two successive reflections is a conformal self-map of the upper half-plane, hence a fractional linear transformation. Each of these is a covering transformation, and in fact, all covering transformations are fractional linear transformations that are compositions of an even number of reflections.



Exercises for XVI.7

1. Show that for $m \geq 1$, the map $\pi(z) = z^m$ is a covering map of $\mathbb{C} \setminus \{0\}$ onto itself.
2. Show that $\pi(z) = e^z$ is a covering map of $\mathbb{C}$ onto $\mathbb{C} \setminus \{0\}$.
3. Let $\pi : S \rightarrow R$ be a covering map. Show that if $\varphi : S \rightarrow S$ is analytic and satisfies $\pi \circ \varphi = \pi$, then φ is one-to-one and onto.
Remark. Thus φ is a covering transformation.

4. Show that if $\{\varphi_n\}$ is a sequence of covering transformations that converges normally to φ , then $\varphi_n = \varphi$ for large n . *Remark.* The covering transformations form a “discrete group.”
5. Consider the covering map $\pi : \mathbb{H} \rightarrow \mathbb{C} \setminus \{0, 1\}$ discussed above. Let V be the domain consisting of $z \in \mathbb{H}$ satisfying $|\operatorname{Re} z| < 1$, $|z - \frac{1}{2}| > \frac{1}{2}$, and $|z + \frac{1}{2}| > \frac{1}{2}$.
 - (a) Show that the images of $V \cup \partial V$ under the covering transformations fill out $\mathbb{H}$. Sketch how the twice-punctured plane $\mathbb{C} \setminus \{0, 1\}$ can be obtained from V by identifying points of ∂V .
 - (b) Identify the covering transformations corresponding to reflection in the imaginary axis followed by reflection in one of the semicircles in ∂V . *Ans.* $w = z/(1 \pm 2z)$.
 - (c) Show that each covering transformation is a finite composition of the four covering transformations $w = z \pm 2$ and $w = z/(1 \pm 2z)$.
 - (d) Show that the covering transformations form a subgroup of the modular group of index 6. *Strategy.* Show that the fundamental domain V is filled out by six congruent copies of the fundamental domain U for the modular group. (See Exercise 1.14.)
 - (e) Show that a unimodular matrix corresponds to a covering transformation for π if and only if the matrix is congruent to the identity matrix modulo 2. *Remark.* The covering transformations form the **principal congruence subgroup** of the modular group of level 2.
6. Let $\pi : S \rightarrow R$ be a covering map, let $q_0 \in S$, and let $p_0 = \pi(q_0) \in R$.
 - (a) Show that if $\gamma(t)$, $0 \leq t \leq 1$, is a path in R from $\gamma(0) = p_0$ to $\gamma(1) = p$, then there is a unique path $\sigma(t)$, $0 \leq t \leq 1$, in S starting at $\sigma(0) = q_0$, such that $\gamma = \pi \circ \sigma$. *Remark.* We say that σ is the **lift** of γ from R to S starting at q_0 .
 - (b) Let γ_0 and γ_1 be two paths in R from p_0 to p_1 , and suppose there is a continuous deformation γ_s , $0 \leq s \leq 1$, of γ_0 to γ_1 with endpoints fixed. Let σ_s be the lift of γ_s to S starting at q_0 . Show that the paths σ_s all terminate at the same point of S , and that σ_s , $0 \leq s \leq 1$, is a continuous deformation of σ_0 to σ_1 .
7. Let γ be the closed path in $\mathbb{C} \setminus \{0, 1\}$ starting at $\frac{1}{2}$ and looping once around the circle $|z| = \frac{1}{2}$ counterclockwise. Let α be the closed path in $\mathbb{C} \setminus \{0, 1\}$ starting at $\frac{1}{2}$ and looping once around the circle $|z - 1| = \frac{1}{2}$ counterclockwise. Let $\alpha\gamma$ denote the path α followed by γ , and similarly for $\gamma\alpha$. Sketch the lifts of each of the four paths α , γ , $\alpha\gamma$, $\gamma\alpha$ to paths in $\mathbb{H}$ starting at the lift $\frac{1}{2} + \frac{1}{2}i$ of $\frac{1}{2}$. Use the result to show that $\alpha\gamma$ cannot be continuously deformed to $\gamma\alpha$.

8. Let $\pi : S \rightarrow R$ be a covering map, let $q_0 \in S$, and let $p_0 = \pi(q_0) \in R$. Show that if $f : \mathbb{D} \rightarrow R$ is an analytic map such that $f(0) = p_0$, then there is a unique analytic map $g : \mathbb{D} \rightarrow S$ such that $g(0) = q_0$ and $f = \pi \circ g$. *Remark.* We say that g is a **lift** of f .
9. Use the covering map $\pi : \mathbb{H} \rightarrow \mathbb{C} \setminus \{0, 1\}$ and the preceding exercise to prove that if $\mathcal{F}$ is a family of analytic functions on the open unit disk $\mathbb{D}$ that omits the points 0 and 1, then $\mathcal{F}$ is a normal family (Montel's theorem). *Strategy.* Let $\{f_n\}$ be a sequence in $\mathcal{F}$, and assume that it has no subsequence that converges normally to 0, 1, or ∞ . Find $z_n \in \mathbb{D}$ with $|z_n| \leq r < 1$ such that the $f(z_n)$'s have lifts ζ_n , $\pi(\zeta_n) = f(z_n)$, that belong to a compact subset of $\mathbb{H}$. Consider the lifts $g_n : \mathbb{D} \rightarrow \mathbb{H}$ of f_n that satisfy $g_n(z_n) = \zeta_n$.
10. Let R be a Riemann surface, and fix a point $p_0 \in R$. Let S be the set of pairs $(p, [\gamma])$, where $p \in R$ and $[\gamma]$ is a homotopy class of paths in R from p_0 to p . If $p \in R$, U_α is a coordinate disk at p , and γ is a path in R for p_0 to p , define $V_{\alpha, \gamma}$ to be the pairs $(q, [\sigma]) \in S$ such that $q \in U_\alpha$ and σ is the path in R that follows γ from p_0 to p and then any path in U_α from p to q . Define $\pi : S \rightarrow R$ to be the coordinate projection $\pi(q, [\gamma]) = q$. (a) Show that S can be made into a Riemann surface in which the sets $V_{\alpha, \gamma}$ are coordinate disks and π is analytic. (b) Show that π is a covering map of S onto R . (c) Show that S is simply connected. *Remark.* Thus S is the universal covering surface of R .

Hints and Solutions for Selected Exercises

Chapter I: The Complex Plane and Elementary Functions

Section I.1

1. (a) circle, (b) annulus, (c) disk, (d) one point, (e) half-plane, (f) horizontal strip, (g) vertical strip, (h) $\mathbb{C} \setminus \mathbb{R}$, (i) half-plane, (j) empty set.
2. (a) Substitute $z = x + iy$, $w = u + iv$, and use the definitions.
3. Set $z = x + iy$, $a = \alpha + i\beta$, then $\operatorname{Re}(\bar{a}z) = \alpha x + \beta y$, and the equation becomes $(x - \alpha)^2 + (y - \beta)^2 = \rho^2$.
4. Apply triangle inequality to $z = \operatorname{Re} z + i \operatorname{Im} z$, to obtain $|z| \leq |\operatorname{Re} z| + |\operatorname{Im} z|$. Equality holds only when z is real or pure imaginary.
6. If $|z| = 1$, then $|z - a| = |z - a||\bar{z}| = |z\bar{z} - a\bar{z}| = |1 - a\bar{z}| = |1 - \bar{a}z|$.
8. Write $p(z) = a_n z^n + \cdots + a_0$ and $h(z) = b_{n-1} z^{n-1} + \cdots + b_0$. Equate coefficients in the polynomial identity $p(z) = (z - z_0)h(z) + p(z_0)$, and solve for the b_j 's in terms of the a_j 's.
11. If n is even, $h(z) = z^{n-2} + z^{n-4} + \cdots + 1$, $r(z) = 1$. If n is odd, $h(z) = z^{n-2} + z^{n-4} + \cdots + z$, $r(z) = z$.

Section I.2

1. (a) $e^{i\pi/4} = (1 + i)/\sqrt{2}$, $e^{5\pi i/4} = -(1 + i)/\sqrt{2}$, (c) $e^{\pm i\pi/4} = (1 \pm i)/\sqrt{2}$, $e^{\pm 3\pi i/4} = (-1 \pm i)/\sqrt{2}$, (e) 2, $2e^{\pm 2\pi i/3} = -1 \pm i\sqrt{3}$, (g) 16.
3. For $0 < b < 1$, an ellipse $x^2/(1+b)^2 + y^2/(1-b)^2 = 1$, traversed in positive direction with increasing θ . For $b = 1$, an interval $[-2, 2]$. For $1 < b < +\infty$, an ellipse traversed in negative direction. For $b = \rho e^{i\varphi}$, express equation as $e^{i\varphi/2}(e^{i(\theta-\varphi/2)} + \rho e^{-i(\theta-\varphi/2)})$ to see that curve is rotate of ellipse or interval by $\varphi/2$.
4. $n = 4, 8, 12, \dots$.
5. (b) Apply (a) to $z = e^{i\theta}$ and to $z = e^{-i\theta}$, add the identities, and use the definitions of sine and cosine.
6. (a) Apply the fundamental theorem of algebra. (b) Expand the product in (a) and find the coefficient of z^{n-1} . (c) Evaluate the identity in (a) at 0. (d) Apply 5(a) to $z = \omega_1^k$.
7. $|z^n + 1| \geq |z^n| - 1 = R^n - 1$, so $1/|z^n + 1| \leq 1/(R^n - 1)$. Now multiply

by $|z^m| = R^m$.

$$8. \cos 4\theta = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta, \sin 4\theta = 4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta$$

Section I.3

2. If $P = (X, Y, Z)$ corresponds to $z = (X + iY)/(1 - Z)$, then $-P$ corresponds to $-(X + iY)/(1 + Z) = -(X^2 + Y^2)/[(1 + Z)(X - iY)] = -(1 - Z^2)/[(1 + Z)(X - iY)] = -(1 - Z)/(X - iY) = -1/\bar{z}$.

4. Show that if z corresponds to (X, Y, Z) , then $1/z$ corresponds to its rotate $(X, -Y, -Z)$.

6. (a) Follows from the fact that the usual euclidean distance in $\mathbb{R}^3$ is a metric. (b) Start with $d(z_1, z_2)^2 = (X_1 - X_2)^2 + (Y_1 - X_2)^2 + (Z_1 - Z_2)^2$, expand, substitute, and do the algebra. (c) $d(z, \infty) = 2/\sqrt{1 + |z|^2}$.

Section I.4

4. (a) Use four sheets, can make branch cuts along real axis from $-\infty$ to 0. (b) Use two sheets, can make branch cuts along horizontal line from $i - \infty$ to i . (c) Use five sheets, can make branch cuts along real axis from $-\infty$ to 1.

Section I.5

3. Check from definitions that $e^{\bar{z}} = e^x(\cos y - i \sin y) = \overline{e^z}$.

4. Substitute $z = 0$, obtain $e^\lambda = 1$, $\lambda = 2\pi mi$.

Section I.6

3. $f(re^{i\theta}) = \log r + i\theta$, $-\pi/2 < \theta < 3\pi/2$.

4. Any straight line cut from z_0 to ∞ , in any direction, will do.

Section I.7

1. (a) $e^{2\pi n} e^{-\pi/4} e^{i \log \sqrt{2}}$, $-\infty < n < \infty$, (b) $-ie^{2\pi n} e^{\pi/2}$, $-\infty < n < \infty$, (c) $\pm 1/\sqrt{2}$, (d) $e^{2\pi n} e^{\log 2 + \pi/3} e^{i(-\log 2 + \pi/3)}$, $-\infty < n < \infty$,

2. $-\pi/2 + i \log 2 + 4\pi m + 2\pi in$, $-\infty < m, n < \infty$,

5. i^i has values $e^{2\pi n} e^{-\pi/2}$, $-\infty < n < \infty$, so that $(i^i)^i$ has values $e^{i(2\pi n - \pi/2 + 2\pi ki)}$, which coincides with $-ie^{2\pi m}$, $-\infty < m < \infty$.

6. Phase factors $e^{2\pi ia}$ at 0 and $e^{2\pi ib}$ at 1. Require $e^{2\pi ia} e^{2\pi ib} = 1$, or $a + b =$ an integer, to have a continuous single-valued determination of $z^a(1 - z)^b$.

7. Use two sheets with slits $[x_1, x_2]$, $[x_3, x_4]$, $\dots$. If n is odd, also need slit $[x_n, +\infty)$. Identify top edge of slit on one sheet with bottom edge of slit on other sheet. Topologically the surface is a sphere with one handle for each slit after the first one, and punctures corresponding to ∞ . If $n = 3$ or $n = 4$, surface is a torus.

8. The function is $z\sqrt{1 - 1/z^3}$. If $|z| > 1$, can use the principal value of the square root to define a branch of the function. There are branch points at $z = 0$ and $z^3 = 1$, that is, at $0, 1, \pm e^{2\pi i/3}$. Make two branch cuts by connecting any two pairs of points by curves; for instance, connect 0 to 1 by a straight line, and the other cube roots of unity by a straight line or arc of unit circle. The resulting two-sheeted surface with identification of cuts and with points at infinity is a torus.

9. Function is $(z+1)\sqrt{z}\sqrt{(1 - 1/z^3)(1 + 1/z)}$. If $|z| > 1$, the second square root can be defined to be single-valued. The values then return to the neg-

ative of their initial values when z traverses the circle, because $\sqrt{z}$ does.

10. Values return to their initial values.

11. Use three sheets. Make two cuts, from 1 to $\pm e^{2\pi i/3}$, on each sheet. In this case the cuts share a common endpoint.

Section I.8

2. Use $\cos(x+iy) = \cos x \cos(iy) - \sin x \sin(iy) = \cos x \cosh y - i \sin x \sinh y$, take the modulus squared, and use $\cosh^2 y = 1 + \sinh^2 y$. The identity for $|\cos z|^2$ shows that the only zeros of $\cos z$ are the zeros of $\cos x$ on the real axis, that is, $\pi/2 + m\pi$, $m = 0, \pm 1, \pm 2, \dots$. Translation by any period λ of $\cos z$ sends zeros to zeros. Thus any period is an integral multiple of π , and since odd integral multiples are not periods, the only periods of $\cos z$ are $2\pi n$, $-\infty < n < \infty$.

3. Since $\cosh z = \cos(iz)$, the zeros of $\cosh z$ are at $i\pi/2 + im\pi$, $m = 0, \pm 1, \pm 2, \dots$, and the periods of $\cosh z$ are $2\pi mi$, $m = 0, \pm 1, \pm 2, \dots$. The function $\sinh z = -i \sin(iz)$ is treated similarly.

4. If $z = \tan w$, then $iz = (e^{iw} - e^{-iw})/(e^{iw} + e^{-iw}) = (e^{2iw} - 1)/(e^{2iw} + 1)$. Solve for e^{2iw} and take logarithms.

6. Use one copy of the doubly slit plane S for each integer n , and define $f_n(z) = \tan^{-1} z + n\pi$ on the n th sheet. Attach the n th sheet to the $(n+1)$ th sheet along one of the cuts, so that $f_n(z)$ and $f_{n+1}(z)$ have the same values at the junction.

Chapter II: Analytic Functions

Section II.1

1. (a) 1, (b) 0, (c) 2, (d) 0.

2. Bounded for $|z| \leq 1$, $\rightarrow 0$ for $|z| < 1$.

4. For fixed k , $(1 - 1/N)(1 - 2/N) \cdots (1 - (k-1)/N) \rightarrow 1$ as $N \rightarrow \infty$.

5. Use $b_n - b_{n-1} = (1/n) - \int_{n-1}^n (1/t) dt < 0$.

9. (a) $\limsup = 2$, $\liminf = 0$, (b) $\limsup = +\infty$, $\liminf = -\infty$, (c) $\limsup = 1$, $\liminf = -1$, (d) $\limsup = +\infty$ if $|x| > 1$, 1 if $|x| = 1$, 0 if $|x| < 1$; $\liminf = -\infty$ if $-\infty < x < -1$, -1 if $x = -1$, 0 if $|x| < 1$, 1 if $x = 1$, $+\infty$ if $x > 1$.

10. (a) continuous on $\mathbb{C}$, (b) continuous for $z \neq 0$, (c) continuous for $z \neq 0$, (d) continuous on $\mathbb{C}$.

11. No limit at points of $(-\infty, 0]$, continuous on $\mathbb{C} \setminus (-\infty, 0]$.

12. No limit at 0, has limit at all other points of closed lower half-plane, limit $= -\pi$ at points of $(-\infty, 0)$.

13. $\rightarrow 0$ if $\operatorname{Re} \alpha > 0$, $\rightarrow 1$ if $\alpha = 0$, otherwise no limit at 0.

15. (a) open, (b) closed, (c) open, (d) neither, (e) open, (f) neither, (g) both open and closed.

Section II.2

1. (a) $2z$, (e) $-2z/(z^2 + 3)^2$, (g) $(ad - bc)/(cz + d)^2$.

2. Differentiate $1 + z + z^2 + \cdots + z^n = (1 - z^{n+1})/(1 - z)$.

4. Use $(f(z + \Delta z) - f(z))/\Delta z = 2az + (bc + 2c\bar{z})\bar{\Delta z}/\Delta z$.

5. Use $(g(z + \Delta z) - g(z))/\Delta z = \overline{(f(\bar{z} + \overline{\Delta z}) - f(\bar{z}))/\overline{\Delta z}}$.
6. Use twice the theorem that if $g_n(t)$ converges uniformly to $g(t)$ for $0 \leq t \leq 1$, then $\int_0^1 g_n(t)dt$ has limit $\int_0^1 g(t)dt$, once to show that $H(z)$ is differentiable, and again to show that the derivative is continuous.

Section II.3

1. (a) $1/\cos^2 z$, (b) $1/\cosh^2 z$, (c) $\tan z \sec z$.
2. $i \cos z$.
5. Use $f' = (\partial u)/(\partial x) - i(\partial u)/(\partial y) = (\partial v)/(\partial y) + i(\partial v)/(\partial x)$.
8. Since ∇v is the rotate of ∇u by $\pi/2$, the directional derivative of v in the θ -direction is equal to the directional derivative of u in the r -direction. This is the first polar Cauchy-Riemann equation. The second follows from the same argument.

Section II.4

2. Write $f(z) = ce^{a \operatorname{Log} z}$ and differentiate.
3. $f'(z) = (1 - 2z)/(2f(z))$.
4. Derivative $= 1/(1 + z^2)$. Any two branches of $\tan^{-1} z$ differ by a constant, so derivatives are same.
5. Derivative $= \pm 1/\sqrt{1 - z^2}$. Derivatives of branches of $\cos^{-1} z$ are not always the same.
7. This is the change of variable formula for a double integral, since the Jacobian is $|f'(z)|^2$.
8. 6π .
9. Use the area formula separately on the top half and the bottom half of the unit disk, where the function is one-to-one. Integral $= 2\pi$.

Section II.5

1. (a) $2xy + C$, (c) $-\cosh x \cos y + C$, (e) $-(1/2)\log(x^2 + y^2) + C$.
4. Use $\partial^2(zh)/\partial x^2 = z\partial^2 h/\partial x^2 + 2\partial h/\partial x$ and similar identity for y -derivatives, add, equate to 0, set $h = u + iv$, and take real and imaginary parts, to obtain Cauchy-Riemann equations for u and v .
7. $\operatorname{Arg} z$ is a harmonic conjugate on $\mathbb{C} \setminus (-\infty, 0]$. Any other harmonic conjugate has form $\operatorname{Arg} z + C$, which does not extend continuously to $\mathbb{C} \setminus \{0\}$.
8. $-(i/2)(\log z)^2$.

Section II.6

1. To aid sketching, express z in polar coordinates.
4. $w = e^{\pi z/(2A)}$.
5. $w = z^{\pi/(2B)}$.
6. Not defined at $z = 0$, not conformal at $z = \pm 1$. If $f(z) = \lambda$, then $z = (\lambda \pm \sqrt{\lambda^2 - 4})/2$. If $z = \rho e^{i\theta}$, then $w = u + iv = \rho \cos \theta + i\rho \sin \theta + (\cos \theta)/\rho - i(\sin \theta)/\rho$, and $u^2/(\rho + 1/\rho)^2 + v^2/(\rho - 1/\rho)^2 = 1$. Since $f(D) = f(\mathbb{D})$, and f is at most two-to-one, f is one-to-one on D . Since f maps $\partial \mathbb{D}$ onto $[-2, 2]$, and f maps onto $\mathbb{C}$, the image of D is $\mathbb{C} \setminus [-2, 2]$.
8. Not defined at $z = 0$, not conformal at $z = \pm e^{i\alpha/2}$. The expression $f(z) = e^{i\alpha/2}(e^{-i\alpha/2}z + 1/e^{-i\alpha/2}z)$ shows that $f(z)$ is a composition of the rotation by $-a/2$, the function of Exercises 6 and 7, and a rotation by $\alpha/2$.

Thus f maps $\{|z| > 1\}$ one-to-one onto the complement rotate of $[-2, 2]$ by $\alpha/2$.

9. Suppose $f(0) = 0$ and $u_x \neq 0$ at 0. The tangents in the orthogonal directions $(1, t)$ and $(-t, 1)$ are mapped to the tangents in the directions $(u_x, v_x) + t(u_y, v_y)$ and $-t(u_x, v_x) + (u_y, v_y)$. The orthogonality of these directions for all t yields $u_x u_y + v_x v_y = 0$ and $u_y^2 + v_y^2 - u_x^2 - v_x^2 = 0$. These equations and some algebra lead to $u_x = \pm v_y$ and $u_y = \mp v_x$.

Section II.7

1. (a) $w = 2i(z - (1 + i))/(z - 2)$, (c) $w = (i - 1)/(z - 2)$, (f) $w = z/(z - i)$, (h) $w = i(z - i)/(z + i)$.
2. Unit circle $\rightarrow$ straight line through 0 and $i - 1$, unit disk $\rightarrow$ half-plane on lower-left of line, real axis $\rightarrow$ straight line through $i - 1$ orthogonal to image of unit circle.
4. Unit circle $\rightarrow$ imaginary axis, unit disk $\rightarrow$ right half-plane, $[-1, 1] \rightarrow$ arc of circle $|w - i/2| = 3/2$ in right half-plane from $-i$ to $2i$.
8. If $w/(\alpha z + \beta)/(\gamma z + \delta)$, divide each coefficient by $\alpha\delta - \beta\gamma$ to obtain representation with $ad - bc = 1$. Representation is not unique, as can multiply all coefficients by -1 .
9. If $f(z)$ maps the three real numbers x_1, x_2, x_3 to $0, 1, \infty$, then $f(z)$ is represented explicitly as $A(z - x_1)/(z - x_2)$, where A is real. Any representation with $ad - bc = 1$ is obtained from this by multiplying each coefficient by the same (real) constant.

Chapter III: Line Integrals and Harmonic Functions

Section III.1

1. (a) $72/5$, (b) 16 , (c) 32 .
3. $2/3$.
4. $-\pi R^2/2$.
6. Differentiate by hand, use uniform convergence of $1/[(z - w)(z - (w + \Delta w))]$ to $1/(z - w)^2$ for $z \in \gamma$.

Section III.2

1. (a) $h = (x^2 + y^2)/2$, (b) $h = (2x^2 + y^6)/6$, (c) $h = xy$, (d) not independent of path, $\int_{|z|=1} = -2\pi$.
2. $\int_{|z|=r} = 2\pi/r^2 \neq 0$.
6. Suppose $|\gamma(t)| = 1$ for $0 \leq t \leq 1$, $\gamma(0) = \gamma(1) = 1$. Follow hint, write $\gamma(t) = e^{i\theta_j(t)}$, $t_{j-1} \leq t \leq t_j$. Note $\theta_j(t_j) - \theta_{j+1}(t_j)$ is an integral multiple of 2π . Add multiples of 2π to the θ_j 's, obtain $\theta(t)$ continuous for $0 \leq t \leq 1$ such that $\theta(0) = 0$ and $\gamma(t) = e^{i\theta(t)}$. Note $\theta(1) = 2\pi m$ for some integer m . Deform by $\gamma_s(t) = e^{i[(1-s)\theta(t) + 2\pi sm]}$.

Section III.3

1. (a) $du = dx - dy$, $dv = dx + dy$, $v = x + y$, (b) $v = 3x^2y - y^3$, (c) $v = \cosh x \sin y$, (d) $v = x/(x^2 + y^2)$.
4. u has a harmonic conjugate v_1 on the annulus slit along $(-b, -a)$, and also a harmonic conjugate v_2 on the annulus slit along (a, b) . Since $v_1 - v_2$

is constant above the slit $(-b, -a)$, and also constant below the slit, v_1 jumps by a constant across the slit. $\text{Arg } z$ also jumps by a constant across the slit. By appropriate choice of C , $v_1 - C \text{Arg } z$ is continuous across the slit $(-b, -a)$, and $u - C \log |z|$ has a harmonic conjugate $v_1 - C \text{Arg } z$ on the annulus. For the identity, use the polar form of the Cauchy-Riemann equations to convert the r -derivative of u to a θ -derivative of v .

Section III.4

1. Express $dx dy$ in polar coordinates centered at z_0 and integrate first with respect to θ .

Section III.5

1. Note that if u attains its maximum or minimum on D , it is constant.
2. $|z^n + \lambda| \leq r^n + \rho$, with equality at $re^{i\varphi/n}e^{2\pi ik/n}$.
6. Set $g = (z + 1)^\varepsilon f$. Then $|g| \leq |f|$. Take R large so that $|g(z)| \leq M$ for $|z| \geq R$. The lim sup condition implies that there is $\delta > 0$ such that $|g(z)| \leq M + \varepsilon$ for $|z| < R$, $0 < \text{Re } z \leq \delta$. Apply the maximum principle to the domain $\{|z| < R, \text{Re } z > \delta\}$, to obtain $|g(z)| \leq M + \varepsilon$. Then let $\varepsilon \rightarrow 0$.
9. Let $\varepsilon > 0$. Take $\delta > 0$ such that $u(z) \geq -\varepsilon$ for $z \in D$, $0 < z < \delta$. Let $\rho > 0$. Take $R > 0$ so large that $u(z) + \rho \log |z| > 0$ for $|z| > R$. By the maximum principle, $u(z) + \rho \log |z| \geq -\varepsilon + \rho \log \delta$ for $z \in D$, $\delta < |z| < R$, hence for all $z \in D$ such that $|z| > \delta$. Let $\rho \rightarrow 0$, then let $\varepsilon \rightarrow 0$, to obtain $u \geq 0$ on D .

Section III.6

1. $\phi = 2x + y$, $\psi = 2y - x$, $f(z) = (2 - i)z$. Streamlines are straight lines with slope $1/2$. Flux across $[0, 1]$ is -1 , across $[0, i]$ is 2 .
2. (a) $\phi = \alpha \log r + \beta \theta$, which is defined only locally, or on a slit plane. (b) $\psi = \alpha \theta - \beta \log r$, $f(z) = (\alpha - i\beta) \log z$. (c) Flux is $2\pi\alpha$, the increase of ψ around circle centered at 0 . Origin is source if $\alpha > 0$, sink if $\alpha < 0$.
5. $\psi(z) = A[2 \arg(e^z - 1) - \pi - 1]$, where $A > 0$.
6. Stream function of conjugate flow is $-\phi$, complex velocity potential is $-i(\phi + i\psi)$.
7. Stream function of conjugate flow is $-\arg z$, complex velocity potential is $-i \log z$, speed is $1/|z|$, particles travel faster near 0 .

Section III.7

1. $(T_1 - T_2)(2/\pi)[\text{Arg}(1 + z) - \text{Arg}(1 - z)] + T_2$
2. $\phi = V_1(2/\pi)[\text{Arg}(1 + z^2) - \text{Arg}(1 - z^2)]$
4. $\phi = 50[\text{Arg}(\sin z + 1) - \text{Arg}(\sin z - 1)]$
5. $u = (2/\pi)(T_1 - T_0)x + (T_0 - T_1)/2$
7. $\phi(x, y, z) = cz$
8. $\phi = -1/r$
9. $\mathbf{F} = (2 - n)\mathbf{u}_r/r^{n-1}$

Chapter IV: Complex Integration and Analyticity

Section IV.1

1. (a) $i/2$, (b) $-1/2$, (c) 0.
2. (a) $2\pi i$ for $m = -1$, otherwise 0, (b) $2\pi i$ for $m = 1$, otherwise 0, (c) 2π for $m = 0$, otherwise 0.
3. (a) 0, (b) $2\pi R^m$, (c) $2\pi i R^m$ for $m = 1$, otherwise 0.
4. Substitute $\bar{z} = x - iy$, $dz = dx + idy$, and apply Green's theorem.
6. $|\operatorname{Log} z| = ((\log R)^2 + \theta^2)^{1/2} \leq \sqrt{2} \log R$ for $R > e^\pi$. Apply the ML -estimate, with $M = \sqrt{2}(\log R)/R^2$, $L = 2\pi R$.

Section IV.2

1. The integrals are all independent of path and can be evaluated by finding a primitive. (a) $2\pi^5 i/5$, (b) 0, (c) $i(e^\pi - e^{-\pi})$, (d) 0.
2. In the right half-plane, use $\operatorname{Log} z$ as a primitive. Integral = $\operatorname{Log}(i\pi) - \operatorname{Log}(-i\pi) = i\pi/2 - i(-\pi/2) = \pi i$. In the left half-plane, use $\log r + i\theta$, $0 < \theta < 2\pi$, as a primitive. Integral = $i\pi/2 - i(3\pi/2) = -\pi i$.
3. A primitive is $z^{m+1}/(m+1)$, $m \neq -1$.

Section IV.3

1. Apply Cauchy's theorem and pass to limit. Integrals over vertical sides of rectangles $\rightarrow 0$ as $R \rightarrow \infty$, by the ML -estimate. $\int_{-R}^R \rightarrow \sqrt{2\pi}$, and the integral over other horizontal side $\rightarrow e^{t^2/2} \int_{-\infty}^{\infty} e^{-x^2/2} e^{-ixt} dx$.
3. (a) Follow hint to estimate $\int_{-1}^1$ by $\int_0^\pi |f(e^{i\theta})|^2 d\theta$ and by $\int_\pi^{2\pi} |f(e^{i\theta})|^2 d\theta$. Use $\int_0^{2\pi} |f(e^{i\theta})|^2 d\theta = \int_0^{2\pi} f(e^{i\theta}) \overline{f(e^{i\theta})} d\theta = 2\pi \sum c_k^2$. (b) Write $c_k = a_k + ib_k$ and apply estimate in (a) twice. (c) The double sum is $\int_0^1 f(x)^2 dx$. Estimate this by $\int_{-1}^1 |f(x)|^2 dx$.

Section IV.4

1. (a) $2\pi i$, (b) 0, (c) 0, (d) πi , (e) $2\pi i/[(m-1)!]$ for $m \geq 1$, 0 for $m \leq 0$, (f) $2\pi i$, (g) $\pi i/2$, (h) $-\pi i/4e^2$.
2. A harmonic function is locally the real part of an analytic function, and any derivative of a harmonic function can be expressed in terms of the real part of a some complex derivative of that analytic function.

Section IV.5

1. If $u \leq C$, and $u = \operatorname{Re} f$, then e^f is bounded, hence constant by Liouville's theorem, and f is constant.
2. If f does not attain values in the disk $|w - c| < \varepsilon$, then $1/(f - c)$ is bounded, hence constant by Liouville's theorem, and f is constant.
4. Apply the Cauchy estimates for $f^{(m+1)}(z)$ to a disk $|z - z_0| < R$, and let $R \rightarrow \infty$, to obtain $f^{(m+1)}(z_0) = 0$. If $f(z)$ is a polynomial of degree $\leq m$ such that $f(z)/z^m$ is bounded near 0, then $f(z) = cz^m$.

Section IV.6

1. Rotate and apply result in text.
2. The analyticity of $H(z)$ follows from the theorem in the text. If $|h| \leq M$, take $C = M(b - a)$ and $A = \max(|a|, |b|)$.

Section IV.8

2. By the Leibniz rule and Exercise 1, the $\bar{z}$ -derivative is $bz + 2c\bar{z}$. The function is complex-differentiable at z if and only if $bz + 2c\bar{z} = 0$. If $b = c = 0$, the function is entire. Otherwise this locus is either $\{0\}$ or a straight line through 0, and there is no open set on which the function is analytic.

6. Using the Leibniz rule, obtain $(\partial/\partial\bar{z})(f\bar{g}) = f(\partial/\partial\bar{z})\bar{g} + \bar{g}(\partial/\partial\bar{z})f = f(\partial/\partial\bar{z})\bar{g} = f\bar{g}'$, then apply (8.4).

7. Since the coefficients of the Taylor expansion depend linearly on f , and since any f is a linear combination of the functions $1, z, \bar{z}, z^2, \bar{z}^2, |z|^2$ and an error term $\mathcal{O}(|z|^3)$, it suffices to check the formula for these six functions.

Chapter V: Power Series

Section V.1

6. Note that $x \log x$ and $x(\log x)^2$ are increasing for $x > 1$, so the terms of both series are decreasing. For $\sum 1/(k \log k)$, there are 2^n terms between $k = 2^n + 1$ and $k = 2^{n+1}$, each $\geq 1/(2^{n+1}(n+1) \log 2)$, so these terms have sum $\geq 1/(2(n+1) \log 2)$. Thus the series diverges, by comparison with the harmonic series. The other series is treated similarly, using an upper estimate. These series can be also treated using the integral test.

7. Observe that $\sum_m^n a_k = S_n - S_{m-1}$, where S_n is the n th partial sum of the series.

Section V.2

1. Set $f'_k(x) = 0$, check that $f_k(x)$ attains its maximum $\varepsilon_k = 1/2\sqrt{k}$ when $x^k = \sqrt{k}$. Since $\varepsilon_k \rightarrow 0$, $f_k \rightarrow 0$ uniformly.

2. $g_k(x) \rightarrow g(x)$, where $g(x) = 0$ for $0 \leq x < 1$, $g(1) = 1/2$, and $g(x) = 1$ for $x > 1$. Convergence is uniform on $[1, 1 - \varepsilon]$ for any $\varepsilon > 0$. Not uniform on $[0, 1]$, because limit function is not continuous at 1. Convergence is uniform on $[1 + \varepsilon, +\infty)$ for any $\varepsilon > 0$.

3. For any $\varepsilon > 0$, $f'_k(z) = z^{k-1}$ converges uniformly for $|z| \leq 1 - \varepsilon$.

4. Apply the Weierstrass M -test.

5. Converges for $x \neq 1$.

6. Check that $x^k/(1 + x^{2k})$ is decreasing for $x \geq 1$. Apply Weierstrass M -test, with M_k the value of the summand at $1 + \varepsilon$.

8. Apply the Weierstrass M -test.

9. If it converges uniformly for $|z| < 1$, then it also converges uniformly for $|z| \leq 1$.

Section V.3

1. (a) $1/2$, (b) 6, (c) 1, (d) $5/3$, (e) $1/\sqrt{2}$, (f) $+\infty$, (g) 2, (h) 0, (i) e .

2. (a) $|z - 1| < 1$, (b) all z , (c) $|z - 2| < 1/2$, (d) $|z + i| \leq 1$, (e) $z = 3$, (f) $|z - 2 - i| \leq 1/2$.

3. Neither series converges at $z = 1$, so $R = 1$.

5. Differentiate the geometric series twice, obtain (a) $z/(1 - z)^2$, (b) $-z + z/(1 - z)^2 + 2z^2/(1 - z)^3$.

Section V.4

1. (a) $\sqrt{2}$, (b) $\pi/2$, (c) $\pi/2$, (d) $\sqrt{5}$, (e) 3, (f) 2.
2. Rewrite $f(z)$ as $(z+1)/(z-e^{2\pi i/3})(z+e^{2\pi i/3})$. Singularities of $f(z)$ are at $\pm e^{2\pi i/3}$, and distance from 2 to nearest singularity is $\sqrt{7}$.
3. $\text{Log } z$ extends to be analytic for $|z-(i-2)| < \sqrt{5}$, though the extension does not coincide with $\text{Log } z$ in the part of the disk in the lower half-plane.
4. Near 0 the function coincides with one of the branches of $(1 \pm \sqrt{1-4z})/2$. The radius of convergence of the power series of either branch is $1/4$, which is the distance to the singularity at $1/4$.
6. $\cosh z = \sum_{n=0}^{\infty} z^{2n}/(2n)!$, $\sinh z = \sum_{n=0}^{\infty} z^{2n+1}/(2n+1)!$, $R = \infty$.
7. $\tan^{-1}(z) = z - z^3/3 + z^5/5 - z^7/7 + \cdots$, converges for $|z| < 1$.
10. Use $f^{(n)}(z) = \alpha(\alpha-1)\cdots(\alpha-n+1)(1+z)^{\alpha-n}$ and the formula for the coefficient of z^n . The series reduces to a polynomial for $\alpha = 0, 1, 2, \dots$. Otherwise radius of convergence is 1, which is distance to the singularity at -1 . Can obtain the radius of convergence also from the ratio test.
13. Use power series.

Section V.5

1. (a) $\sum_{n=0}^{\infty} (-1)^n/z^{2n+2}$, (c) $\sum_{n=0}^{\infty} 1/n!z^{2n}$, (d) $\sum_{n=0}^{\infty} 1/(2n+1)!z^{2n}$.
4. If $|z| \leq M$ for $z \in E$, and $R > M$, then $1/(w-z) = \sum z^n/w^{n+1}$ converges uniformly for $z \in E$ and $|w| > R$. Integrate term by term, obtain $f(w) = \sum_{n=0}^{\infty} b_n/w^{n+1}$, $|w| > R$, where $b_n = \iint_E z^n dx dy$.
5. $f(w) = \pi/w$ for $|w| \geq 1$, $f(w) = \pi\bar{w}$ for $|w| \leq 1$. To find the formula for $|w| < 1$, break the integral into two pieces corresponding to $|z| > |w|$ and to $|z| < |w|$, and use geometric series.

Section V.6

1. $1/\cos z = 1 + (1/2)z^2 + (5/24)z^4 + (1/12)z^6 + \mathcal{O}(z^8)$.
2. $z/\sin z = 1 + (1/6)z^2 + (7/360)z^4 + \mathcal{O}(z^6)$.
3. $R = 1 =$ distance to singularity at -1 .
4. $B_1 = 1/6$, $B_2 = B_4 = 1/30$, $B_3 = 1/42$, $B_5 = 5/66$.
5. $E_0 = 1$, $E_2 = -1$, $E_4 = 5$, $E_6 = -61$.
6. $f_m^{(k)}(z) \rightarrow f^{(k)}(z)$ uniformly for $|z| \leq \rho - \varepsilon$, so $a_{k,m} = f_m^{(k)}(0)/k! \rightarrow f^{(k)}(0)/k! = a_k$.

Section V.7

1. (a) simple zeros at $\pm i$, (b) simple zeros at $\pm e^{\pi i/4}, \pm e^{3\pi i/4}$, (c) triple zero at 0, simple zeros at $n\pi$, $n = \pm 1, \pm 2, \dots$, (d) double zeros at $n\pi$, $n = 0, \pm 1, \pm 2, \dots$, (h) simple zeros at $-\pi i/4 + n\pi i/2$, $n = 0, \pm 1, \pm 2, \dots$, (i) no zeros.
2. (a) analytic at ∞ , (b) analytic at ∞ , simple zero, (c)-(i) not analytic at ∞ .
6. Apply argument in text to $f(z) - f(z_0)$.
9. Write $f(z) = (z - z_0)^N h(z)$, where $h(z)$ has a convergent power series and $h(z_0) \neq 0$. Take $g(z) = (z - z_0)e^{(\log h(z))/N}$ for an appropriate branch of the logarithm.
10. Show that the zeros of $f(z)$ are isolated. At a zero of $f(z)$, write

$f(z)^N = (z - z_0)^m h(z)$ where $h(z_0) \neq 0$, and show that N divides m .

13. $f(D) \subset D \cup \partial D$. If $f(z)$ is not constant, then $f(D)$ is open, and $f(D)$ cannot contain any point of ∂D .

Section V.8

1. $\sqrt[3]{z^3 - 1}$ returns to $e^{2\pi i/3}$ times initial value, other functions return to initial values.

2. $f_t(z) = it + \sum_{m=1}^{\infty} ((-1)^{m-1} e^{-itm}/m)(z - e^{it})^m$, $f_{2\pi}(z) = f_0(z) + 2\pi i$.

9. Let t_1 be the infimum of $t > a$ such that the power series expansion of $P(z, f_t(z))$ at $\gamma(t)$ is not identically zero, and apply the uniqueness principle for $t < t_1$ near t_1 .

Chapter VI: Laurent Series and Isolated Singularities

Section VI.1

1. (a) Laurent expansions $-\sum_{n=-1}^{\infty} z^n$ for $0 < |z| < 1$, and $\sum_{n=-\infty}^{-2} z^n$ for $|z| > 1$.

2. (a) $\sum a_n(z+1)^n$, where $a_n = -1$ for $n \leq -1$ and $a_n = -1/2^{n+1}$ for $n \geq 0$. Converges for $1 < |z+1| < 2$.

4. If $f(z) = \sum a_n z^n$ is even, then $f(z) = f(-z) = \sum a_n (-1)^n z^n$. By the uniqueness of the expansion, $a_n = (-1)^n a_n$, and $a_n = 0$ if n is odd.

5. Let $f(z) = \sum a_n z^n$. Then $a_{-1} = \int_{|z|=r} f(z) dz$, and $f(z) - a_{-1}/z$ has the primitive $\sum_{n \neq -1} (a_n/(n+1))z^{n+1}$.

Section VI.2

1. (a) double poles at ± 1 , principal parts $\mp(1/4)/(z \pm 1)^2$, (c) removable singularity at 0, (e) essential singularity at 0, (g) analytic on $\mathbb{C} \setminus [0, 1]$, no isolated singularities, (h) double pole at 1, principal part $1/(z-1)^2 - (1/2)/(z-1)$.

2. (a) 3, (b) ∞ , (c) $\pi\sqrt{2}$, (d) π .

3. (a) $\tan z$ has two poles in the disk $|z| < 4$, simple poles at $\pm\pi/2$, principal parts $1/(z \pm \pi/2)$. If $f_1(z) = 1/(z - \pi/2) + 1/(z + \pi/2)$, then $f_0(z) = f(z) - f_1(z)$ is analytic for $|z| < 4$, and $f_1(z)$ is analytic for $|z| > 3$ and $\rightarrow 0$ as $z \rightarrow \infty$. By uniqueness, $f(z) = f_0(z) + f_1(z)$ is the Laurent decomposition. (b) Use geometric series. Converges for $|z| > \pi/2$. (c) $a_0 = a_2 = 0$, $a_1 = 1 + 8/\pi^2$. (d) $3\pi/2$.

4. Use uniqueness of the Laurent decomposition.

7. By Riemann's theorem, $g(z) = (z - z_0)^N f(z)$ has a removable singularity at z_0 , hence $f(z) = g(z)/(z - z_0)^N$ is meromorphic at z_0 .

11. For starters, check the function $f(z) = 1/(z - z_0)^N$.

13. Suppose values of $f(z)$ do not cluster at L as $z \rightarrow z_0$. Then $g(z) = 1/(f(z) - L)$ is bounded for $|z - z_0| < \varepsilon$, $z \neq z_j$. Apply Riemann's theorem first to the z_j 's for j large, then to z_0 , to see that $g(z)$ extends to be analytic for $|z - z_0| < \varepsilon$, and $f(z)$ is meromorphic there.

Section VI.3

1. (a) removable singularity at ∞ , (c) essential singularity at ∞ , (e) simple pole at ∞ , (g) removable singularity at ∞ , (h) ∞ not an isolated singularity.

2. Laurent expansion has infinitely many negative powers of $1/z$, so singularity at ∞ is essential.
3. If e^f has a removable singularity at ∞ , then it is bounded, hence constant by Liouville's theorem, and f is constant. If e^f has a pole at ∞ , then e^{-f} is bounded, hence constant by Liouville's theorem, and f is constant.
4. (c) Use binomial series to expand each branch of $f(z) = z(1 - 1/z^3)^{1/2}$ in a Laurent series at ∞ . Each branch is analytic for $|z| > 1$ and has a simple pole at ∞ .

Section VI.4

1. (b) $-1/z + 1/(z-1) + 1/(z+1)$ (d) $(-1/4)/(z-i)^2 + (-i/4)/(z-i) + (-1/4)/(z+i)^2 + (i/4)/(z+i)$ (f) $1 - 3/(z+2)$.
2. (a) $z - [(1+i)/2]/(z-i) + [(-1+i)/2]/(z+i)$ (b) $z^3 - (1/3)/(z-1) + (\omega/3)/(z-\omega) - (\omega/3)/(z+\omega)$, where $\omega = e^{2\pi i/3}$.
3. Dimension = 5, with basis $1, 1/z, 1/z^2, 1/(z-i), 1/(z-i)^2$.

Section VI.5

2. $1/\cos(2\pi z) = 2e^{2\pi iz}/(1 + e^{4\pi iz}) = 2e^{2\pi iz} \sum_{m=0}^{\infty} (-1)^m e^{4m\pi iz}$, converges absolutely for $\text{Im } z > 0$. For any $\varepsilon > 0$, converges uniformly for $\text{Im } z \geq \varepsilon$.
5. If ω is a period $\neq 0$ or ± 1 , then $|\omega \pm 1| \geq 1$. If moreover $\omega \neq \pm i$, then $|\omega^2 + 1| \geq 1$. The only possibilities for periods on unit circle are then $\{\pm 1\}$, $\{\pm 1, \pm i\}$, and $\{\pm 1, \pm e^{\pi i/3}, \pm e^{2\pi i/3}\}$.

Section VI.6

2. $\theta \sim \sum_{k=-\infty}^{\infty} c_k e^{ik\theta}$, $c_k = i(-1)^k/k$ for $k \geq 1$, $c_k = -i(-1)^k/k$ for $k \leq -1$, $c_0 = 0$. The terms of the differentiated series do not tend to 0, so the differentiated series diverges at each θ . At $\theta = \pi$, the complex Fourier series for $k \geq 1$ becomes a multiple of the harmonic series, hence the series diverges at $\theta = \pi$. The corresponding sine series is $\theta \sim \sum_{k=1}^{\infty} b_k \sin(k\theta)$, $b_k = 2(-1)^{k+1}/k$. It converges to θ if $-\pi < \theta < \pi$ and to 0 if $\theta = \pm\pi$.
3. The cosine series is $\sum a_k e^{ik\theta}$, where $a_0 = \pi^2/3$, and $a_k = 4(-1)^k/k^2$ for $k \geq 1$. By Weierstrass M -test, series converges uniformly for $|\theta| \leq \pi$.
5. Since $|c_k e^{ik\theta}| = |c_k|$, series converges absolutely for all θ . By Weierstrass M -test, convergence is uniform in θ .

Chapter VII: The Residue Calculus

Section VII.1

1. (a) $-i/4$, (b) $i/4$, (e) 0, (i) $2e^{2\pi ik/n}/n$.
2. (a) 1 at $z = 0$, (b) -1 at $z = (\pi/2) + n\pi$.
3. (a) residue at $z = 0$ is 1, integral = $2\pi i$, (f) $z = 0$ is removable, pole inside contour at $z = \pi/2$, residue = -1 , $\int = -2\pi i$.
4. Apply Rule 3 to evaluate the residues.
5. The integrals along opposite sides cancel.
6. (a) pole at $z = 0$, residue = $e^{\pi i/4}/2\pi i = (1-i)/2\sqrt{2}\pi$.

Section VII.2

1. Use semicircular contour, residue at ia is $1/2ia$.
3. Use Rule 2 to find residue $1/4i$ at $z = i$.

5. Use Rule 3 to find residues at $e^{\pi i/4}$ and $e^{3\pi i/4}$.
9. Use semicircular contour to show $\int_{-\infty}^{\infty} [e^{2ix}/(x^2+1)]dx = \pi e^{-2}$. Residue of $e^{2iz}/(z^2+1)$ at i is $e^{-2}/\pi i$. Use also $-4\sin^2 x = (e^{ix} - e^{-ix})^2 = e^{2ix} + e^{-2ix} - 2$.
10. Integral exists only for a real, and $\operatorname{Re} b \neq 0$ or $\cos(iab) = 0$. It depends analytically on b for $\operatorname{Re} b \neq 0$.

Section VII.3

1. Poles of $(z^2+1)/[z(z^2+4z+1)]$ are at 0 and $-2 \pm \sqrt{3}$. Use Rule 3 to obtain residue 0 at 1 and residue $-2 + \sqrt{3}$ at $-2\sqrt{3}/3$. Other pole is outside the unit circle.
4. Poles of $z/(z^4 - 6z^2 + 1)$ are at $z_0 = \pm\sqrt{3-2\sqrt{2}}$, residue by Rule 3 is $1/(4z_0^2 - 12) = -1/8\sqrt{2}$.
7. Choose branch of $\sqrt{w^2-1}$ on $\mathbb{C} \setminus [-1, 1]$ that is positive for $w \in (1, \infty)$. It suffices to check the identity for $w = a > 1$. Poles of $z/(z^2 + 2az + 1)^2$ are double poles at $z_{\pm} = -a \pm \sqrt{a^2-1}$. Use Rule 2 to obtain residue at z_+ of $-(z_+ + z_-)/(z_+ - z_-)^3 = a/4(a^2-1)^{3/2}$.

Section VII.4

2. Simple pole at $e^{i\pi/b}$ with residue $-e^{i\pi/b}/b$. Use $z = re^{2\pi i/b}$, $dz = e^{2\pi i/b}dr$, to parametrize one edge of the domain. Obtain $(1 - e^{2\pi i/b}) \int_0^{\infty} = -2\pi i e^{i\pi/b}/b$.
4. Integrate $z^{-a}/(1+z)^m$ around the keyhole contour. Specify branch by $z^{-a} = r^{-a}e^{-ia\theta}$, $0 < \theta < 2\pi$. Determine the residue from power series expansion of z^{-a} about -1 .
7. This is similar to Exercise 2. The integral depends analytically on the parameter a for $0 < \operatorname{Re} a < \operatorname{Re} b$.
9. Integrate $(\log z)^2/(z^3+1)$ around a pie-slice domain with $0 < \theta < 2\pi/3$. Specify branch by $\log z = \log r + i\theta$, $0 < \theta < 2\pi/3$. Simple pole at $e^{\pi i/3}$, residue $-\pi^2 e^{-2\pi i/3}/27$. Apply residue theorem, pass to limit, multiply by $e^{-\pi i/3}$, take imaginary parts, and substitute the values for the integrals from Exercise 8, to obtain $-\sqrt{3} \int_0^{\infty} + 4\pi^3/81 + 12\pi^3/81 = 2\pi^3/27$. Then solve for the integral.

Section VII.5

1. Specify branch by $f(z) = r^{-a}e^{-ia\theta}(\log r + i\theta)/(z-1)$, $z = re^{i\theta}$, $0 < \theta < 2\pi$. It is analytic on the keyhole domain. It extends analytically to $(0, \infty)$ from above, and the apparent singularity at $z = 1$ is removable. However, the extension to $(0, \infty)$ from below has a simple pole at $z = 1$, with residue $2\pi i e^{-2\pi ia}$ obtained by using $\theta = 2\pi$ in the definition of $f(z)$. Use Cauchy's theorem, multiply by $e^{2\pi ia}$, take real parts, and pass to limit using the fractional residue formula with angle $-\pi$.
3. For $a = 1$, integrate $e^{iz}/[z(\pi^2 - z^2)]$ around semicircular contour indented at 0 and $\pm\pi$. Take imaginary parts, then pass to limit using fractional residue theorem. Residue at 0 is $1/\pi^2$, at $\pm\pi$ is $1/2\pi^2$.
5. Integrate around semicircular contour indented at 0 and $\pm\pi$. Take real parts and pass to limit, using $\operatorname{Re}(e^{2ix} - 1) = -2\sin^2 x$. Residue at 0 is $2i$.

Fractional residue theorem gives limit $(-\pi i)2i = 2\pi$ for the integral over the indentation.

Section VII.6

1. $\int_0^{1-\varepsilon} + \int_{1+\varepsilon}^{\infty} = (1/2) \log[(2-\varepsilon)/(2+\varepsilon)] \rightarrow 0$ as $\varepsilon \rightarrow 0$.
3. By residue theorem, $\int_{-\infty}^{a-\varepsilon} + \int_{C_\varepsilon} + \int_{a+\varepsilon}^{\infty} = \pi/(i-a) = -\pi(i+a)/(a^2+1)$. By the fractional residue theorem, $\int_{C_\varepsilon} \rightarrow -\pi i/(a^2+1)$.
5. There are two fractional residues, at 1 and at $e^{2\pi i/b}$, each with angle $-\pi$. By Cauchy's theorem, $(1 - e^{2\pi i/b} e^{2\pi(a-1)i/b}) \text{PV} \int -\pi i \text{Res}[1] - \pi i \text{Res}[e^{2\pi i/b}] = 0$. The residue at 1 is $e^{2\pi i/b}$, at $e^{2\pi i/b}$ is $e^{2\pi i/b} e^{2\pi(a-1)i/b}/b$.

Section VII.7

2. Integrate $z^3 e^{iz}/(z^2+1)^2$ around semicircular contour of radius R in the upper half-plane. Residue at i is $1/4e$, integral $= \pi i/2e$. Let $R \rightarrow \infty$, use Jordan's lemma, and take the imaginary part.
3. The limit is πe^{-a} if $a > 0$, and $-\pi e^a$ if $a < 0$. For $a > 0$, integrate $z e^{iaz}/(z^2+1)$ around semicircular contour. Residue at i is $e^{-a}/2$. Pass to limit, use Jordan's lemma, and take the imaginary part.

Section VII.8

1. (a) -1 , (b) 0 , (c) -1 , (d) 0 , (e) $-1/(n+1)!$, (f) $\pm(b-a)/2$.
2. Follow calculation in text, use coefficient of $1/z^5$ in binomial series, residue at ∞ is $-35i/128$.
3. Converges only if $n \geq 0$, for $0 < \text{Re } a < n+1$. Converges to $\pi/\sin(\pi a)$ if $n = 0$, to $(-1)^n[(a-1) \cdots (a-n)/n!]\pi/\sin(\pi a)$ if $n > 0$. Evaluate it for $0 < a < n+1$ by integrating $z^n/z^a(1-z)^{1-a}$ around dogbone contour.
5. Consider $\int f(z)dz$ around a large circle.
7. Residue at ∞ is $-3/8$, so sum of residues in the finite plane is $3/8$.

Chapter VIII: The Logarithmic Integral

Section VIII.1

2. Two in second quadrant, two in third.
5. Four zeros in open left half-plane for $\alpha < 1$ and $\alpha > 3$, two for $1 \leq \alpha \leq 3$. For $\alpha = 1$ and $\alpha = 3$ there are also two zeros on the imaginary axis.
6. Two zeros in open right half-plane for $\alpha \leq 0$, one zero for $\alpha = 1$, and three zeros for $\alpha > 0$, $\alpha \neq 1$.
8. Increase in argument of $z + \lambda - e^z$ around large rectangle with vertices $\pm iR$ and $-R \pm iR$ is 2π , so there is one zero in rectangle.
9. There is an analytic branch $\text{Log } f(z)$ of $\log f(z)$ on D , so the increase in $\arg f(z)$ around any closed path in D is zero.

Section VIII.2

1. On circle $|z| = 1$ take BIG $= 6z$. On circle $|z| = 2$ take BIG $= 2z^5$.
2. Six.
3. Use $|p(z) - 3z^n| < e$ for $|z| = 1$.
4. Use $|e^z| \geq 1$ for $\text{Re } z > 0$. The zeros are simple unless $\lambda = 0$.
6. (a) One in each quadrant. (b) Second and third quadrants. (c) Compare

$p(z)$ with $z^6 + 9z^4$.

7. Neither $f(z)$ nor $g(z)$ has zeros on ∂D , so each has at most finitely many zeros in D . Estimate shows $f(z)/g(z) \notin (-\infty, 0]$ for z near ∂D , so $\text{Log}(f(z)/g(z))$ is continuously defined near ∂D . Increase in $\arg f(z)/g(z)$ around any closed path near ∂D is zero.

8. They have the same number of zeros minus poles.

9. If $A = (\partial f / \partial \bar{z})(z_0) \neq 0$, consider Taylor approximation of $f(z)$ at z_0 , show increase in argument of $A(\bar{z} - \bar{z}_0) + o(|z - z_0|)$ around $|z - z_0| = \varepsilon$ is < 0 .

Section VIII.3

1. Take $\text{BIG} = f(z)$, little $= f_k(z) - f(z)$.

Section VIII.4

1. Apply the residue theorem. Residue at z_j is $m_j g(z_j)$.

4. Suppose that $f(z)$ is analytic across $\partial \mathbb{D}$, and consider the set of $z \in \mathbb{D}$ whose image $f(z)$ is not the image of any other point in $\mathbb{D}$.

5. (a) Apply the argument principle to $f(z) - w$ on a large disk.

6. Choose w_0 such that the number of points in $f^{-1}(w_0)$ is maximum. Then $f(z)$ attains values w near w_0 only near points in $f^{-1}(w_0)$, $1/(f(z) - w_0)$ is bounded at ∞ , and $f(z)$ is meromorphic on $\mathbb{C}^*$ hence rational.

8. Let $w_0 \in \partial f(D)$, and take $z_n \in D$ with $f(z_n) \rightarrow w_0$. Assume $z_n \rightarrow z_0 \in D \cup \partial D$, then $f(z_0) = w_0$. Since $f(D)$ is open, $w_0 \notin f(D)$. Thus $z_0 \in \partial D$, and $w_0 \in f(\partial D)$.

Section VIII.5

1. Critical points ± 1 , critical values ± 2 .

3. Critical point 0, critical value 1.

5. Its derivative is a polynomial of degree $m - 1$, hence has $m - 1$ zeros counting multiplicity.

7. Denote the k th iterate by $f_k(z)$, a polynomial of degree m^k . By the chain rule, $f'_N(z) = f'(f_{N-1}(z))f'(f_{N-2}(z)) \cdots f'(f(z))f'(z)$. Thus the critical points of $f_N(z)$ consist of the $m^k(m - 1)$ inverse images of the critical points of $f(z)$ under $f_k(z)$ for $0 \leq k < N$. Total number of critical points is $m^N - 1$.

10. Critical point at 0, ± 1 of order 1 and at ∞ of order 3. Critical values are 0, -1 , ∞ .

13. $z_1(w) = e^{iw}$, $z_2(w) = e^{-iw}$, graphs meet at $w = \pi m$, $-\infty < m < \infty$.

Section VIII.6

1. $W(\gamma, \zeta) = 1$ for ζ in the four bounded components of $\mathbb{C} \setminus \gamma$, $W(\gamma, \zeta) = 1$ for ζ in the unbounded component.

4. (a) If γ is a closed path in D , then $W(\gamma, z)$ is constant on any connected set disjoint from γ , so $W(\gamma, z_0) = W(\gamma, z_1) = m$, and the increase in the argument of $f(z)$ is $4\pi m$. (b) The increase in the argument of $\sqrt{f(z)}$ around γ is $2\pi m$, so the analytic continuation of a branch of $\sqrt{f(z)}$ around any closed path in D returns to itself, and we can define an analytic branch of $\sqrt{f(z)}$ by analytic continuation.

5. For $n \geq 2$, $(z - z_0)^{-n}$ has primitive $(z - z_0)^{-n+1}/(1 - n)$ on D .
8. Apply the theorems in III.2 on deformation of paths.
9. The increase in $\arg f(z)$ around the circle $|z| = r$ is constant for $1 \leq r < \infty$, and it tends to 0 as $r \rightarrow \infty$, so it is identically zero. The argument principle, applied to $f(z)$ on $\mathbb{D}$, shows that $f(z)$ has no zeros in $\mathbb{D}$.
10. Suppose $0 \notin f(K)$ and $f(\infty) \neq 0$. The increase in $\arg f(z)$ around the boundary of a large square S is then 0, and $f(z)$ has only finitely many zeros. Cut S into a grid of very small squares S_j and add the increases of $\arg f(z)$ over ∂S_j .

Section VIII.7

1. As ζ crosses the curve from left to right, the Cauchy integral jumps by $-2\pi i f(\zeta)$, so the value of the integral on the lower half-plane is $g(\zeta) - 2\pi i f(\zeta)$.
2. Both sides are analytic on $\mathbb{C} \setminus [-1, 1]$. They are equal for $\zeta \in (-\infty, -1)$, hence they coincide on $\mathbb{C} \setminus [-1, 1]$. This does not contradict Exercise 1, since the logarithm function is not analytic on $(-1, 1)$.
3. (a) $F(\zeta) = 0$ for $|\zeta| > 1$, $F(\zeta) = \zeta$ for $|\zeta| < 1$. (b) $F(\zeta) = -1/\zeta$ for $|\zeta| > 1$, $F(\zeta) = 0$ for $|\zeta| < 1$. (c) $F(\zeta) = -1/2\zeta$ for $|\zeta| > 1$, $F(\zeta) = \zeta/2$ for $|\zeta| < 1$. (d) $F(\zeta) = 1/2i\zeta$ for $|\zeta| > 1$, $F(\zeta) = \zeta/2i$ for $|\zeta| < 1$.

Section VIII.8

1. (a) and (c) are simply connected, (b) and (d) not.
3. (a) is simply connected, (b) is not.
5. If D is simply connected, and if $G(z)$ is a branch of $\log f(z)$, then $e^{G(z)/2}$ is a branch of $\sqrt{f(z)}$. If D is not simply connected, and if z_0 and K are as in the proof that (iv) $\Rightarrow$ (v), then $z - z_0$ cannot have an analytic square root $g(z)$, or else $\int_{\partial K} d \arg g(z) = \pi$ and not an integer times 2π .
9. Since $f(D)$ is simply connected, the analytic continuation of $g(w)$ is independent of path, by the monodromy theorem. Hence there is an analytic function $g(w)$ defined on $f(D)$ such that $g(f(z)) = z$ for z near z_0 . By the uniqueness principle, $g(f(z)) = z$ for all $z \in D$. This implies $f(z)$ is one-to-one on D .
10. Assume $h(\zeta)$ attains only the values 0 and 1, let $E = h^{-1}(1)$, and consider $\sigma = \partial K$ as in the proof that (iv) $\Rightarrow$ (v).
11. $W(\sigma, \zeta) = (1/2\pi i) \int_{\partial U} 1/(z - \zeta) dz = 0$ for $\zeta \notin D$, by Cauchy's theorem.
12. Let E be a bounded component of $\mathbb{C} \setminus D$. Modify ∂K constructed in the proof that (iv) $\Rightarrow$ (v) to obtain a single closed curve γ such that $W(\gamma, \zeta) = 1$ if $\zeta \in E$ and $W(\gamma, \zeta) = 0$ if $\zeta \notin D \cup E$.

Chapter IX: The Schwarz Lemma and Hyperbolic Geometry

Section IX.1

3. Since $f(z) = 1$ has a solution $z_1 = 0$, the argument principle guarantees that $f(z) = 0$ also has a solution z_0 , $|z_0| < 1$. If $\varphi(w) = M(w-1)/(M^2-w)$, then $|\varphi(f(z))| < 1$ for $|z| = 1$, and $\varphi(f(0)) = 0$. By the Schwarz lemma, $1/M = |\varphi(f(z_0))| < |z_0|$.

5. Assume $f(0) = r$, and apply the Schwarz lemma to $\psi \circ f$, where $\psi(\zeta) = (\zeta - r)/(1 - r\bar{\zeta})$. Equality holds at z_0 only when $f(z) = \lambda(\mu z + r)/(1 + r\bar{\mu}z)$, where $|\lambda| = |\mu| = 1$, and $z_0 = -\bar{\mu}s$, $0 \leq s < r$.
6. If $f(0)$ has distance d from ∂D , then the inverse function $f^{-1}(w)$ is analytic for $|w - f(0)| < d$, and $|f^{-1}(w)| < 1$. Apply the Schwarz lemma to obtain $|(f^{-1})'(f(0))| = 1/|f'(0)| \leq 1/d$.
8. $f(z)/z$ is analytic for $|z| < 1$, and $|f(z)/z| < 1$ for $|z| < 1$, or else $f(z)/z$ would be a unimodular constant, violating $|f'(0)| < 1$. Take c to be the maximum of $|f(z)/z|$ for $|z| \leq r$.

Section IX.2

1. Let $B(z)$ be the finite Blaschke product with same zeros as $f(z)$. Then $g(z) = f(z)/B(z)$ is continuous for $|z| \leq 1$, analytic for $|z| < 1$, and satisfies $|g(z)| = 1$ for $|z| = 1$. Hence $|g(z)| \leq 1$ for $|z| \leq 1$. Since $g(z)$ has no zeros, the maximum principle applies also to $1/g(z)$, and $|g(z)| \geq 1$. Hence $g(z)$ is constant.
3. Write $f(z/3) = B(z)g(z)$, where $B(z)$ is the Blaschke product with zeros at $\pm i/3$ and $\pm 1/3$. The maximum value is $|B(0)| = 1/81$.
4. Let $d(z_0, z_1)$ denote the maximum. Since the analytic functions can be precomposed with conformal self-maps, $d(z_0, z_1) = d(0, s)$, where $s = |z_0 - z_1|/|1 - \bar{z}_0 z_1|$ is the image of z_1 under conformal self-map sending z_0 to 0. Then $d(0, s) = (2/s)(1 - \sqrt{1 - s^2})$, and the maximum is attained by the conformal self-map that satisfies $g(0) = -g(s)$.
9. By Riemann's theorem on removable singularities, $f(z)$ is analytic at 0. Since $f(z)$ maps the punctured disk one-to-one onto itself, the value $f(0)$ must belong to the boundary of the punctured disk, and it satisfies $|f(0)| < 1$, so $f(0) = 0$. Thus f is a conformal self-map of $\mathbb{D}$ satisfying $f(0) = 0$, and f is a rotation.
12. For (a) there are six, for (b) there are eight, and for (c) there is only the identity $f(z) = z$.
15. (a) The equation $f(z) = z$ becomes $z^2 + ((\lambda - 1)/\bar{a})z - \lambda a/\bar{a} = 0$, whose solutions satisfy $|z_0 z_1| = 1$. (b) If $z_0 \in \mathbb{D}$ is fixed by f , and h is conformal self-map sending z_0 to 0, then 0 is fixed by $h \circ f \circ h^{-1}$. (d) Conjugate f by an h that maps ± 1 to the two fixed points of f . Can also replace $\mathbb{D}$ by $\mathbb{H}$, place the fixed points at 0 and ∞ , and show f is conjugate to $w \rightarrow Aw$ on $\mathbb{H}$. (f) Replace $\mathbb{D}$ by $\mathbb{H}$, place the fixed point at ∞ , and show that f is conjugate to the translation $w \rightarrow w + 1$ on $\mathbb{H}$.

Section IX.3

2. A hyperbolic disk centered at 0 of hyperbolic radius ρ is a Euclidean disk centered at 0 with radius r given by solving $\rho = \log[(1 + r)/(1 - r)]$. Since conformal self-maps are isometries in the hyperbolic metric, they map hyperbolic disks to hyperbolic disks of the same radius, and every hyperbolic disk is the image under a conformal self-map of a hyperbolic disk centered at 0. Conformal self-maps also map Euclidean disks to Euclidean disks.

3. Let r be the Euclidean radius of hyperbolic disk of radius ρ centered at 0. For $0 < s < 1$, this disk is mapped by $(z + s)/(1 + sz)$ to the hyperbolic disk of radius ρ centered at s . The diameter $(-r, r)$ is mapped to the interval from $(s - r)/(1 - sr)$ to $(s + r)/(1 + sr)$. Since the Euclidean center is the midpoint of this interval, this yields explicit formulae for $c(s, \rho)$ and $r(s, \rho)$, from which the limits can be evaluated explicitly.
4. The hyperbolic circumference is $2 \int_{\gamma} |dz|/(1 - |z|^2)$. Set $z = re^{i\theta}$ and integrate, then substitute $r = (e^{\rho} - 1)/(e^{\rho} + 1)$.
7. Consider first an isometry $f(z)$ that fixes two points, $f(0) = 0$ and $f(r) = r$. For $z \in \mathbb{D} \setminus \mathbb{R}$, consider the two hyperbolic circles centered at 0 of radius $\rho(0, z)$ and centered at r of radius $\rho(r, z)$, and observe that they meet at only the two points z and $\bar{z}$.
8. $w = f(z)$ is an isometry if and only if $|dw|/(1 + |w|^2) = |dz|/(1 + |z|^2)$. Use $dw/dz = 1/(cz + d)^2$ to obtain equivalent equations $|a|^2 + |c|^2 = 1 = |b|^2 + |d|^2$, $a\bar{b} + c\bar{d} = 0$.
9. (a) Let γ be the geodesic from z to ζ , and estimate $\rho(z^2, \zeta^2)$ by the hyperbolic length of $f \circ \gamma$. (b) Take ζ close to z .
13. Use $g(z) = (z - i)/(z + i)$ in the definition.
14. Use $g(z) = (e^z - 1)/(e^z + 1)$ in the definition.

Chapter X: Harmonic Functions and the Reflection Principle

Section X.1

3. Use (1.11).
5. Follow the second proof of the boundary-value theorem.
6. Same as Exercise 5, except must use the symmetry of the Poisson kernel to see that each side contributes equally.
8. (a) Justify differentiating under the integral in the Poisson integral formula.

Section X.2

1. Integrate around a circle and interchange the order of integration.
2. Approximate $u(x, y)$ by a Taylor polynomial of degree two. Suffices to check formula for 1, x , y , x^2 , xy , and y^2 .
4. If $r_1/r_2 = \rho_1/\rho_2$, set $a = \rho_1/r_1 = \rho_2/r_2$, $f(z) = e^{ia y}$, and use Exercise 3.

Section X.3

2. Express reflection in circle as composition of two fractional linear transformations and the reflection $z \rightarrow \bar{z}$ in the real axis.
3. If curves meet at angle θ , their reflections meet at angle $-\theta$.
4. (a) Use $z'(\zeta) \neq 0$ at $\zeta = 0$. (b) $\zeta(z)$ maps an interval on γ onto an interval on $\mathbb{R}$. (c) By definition, $z^* = \bar{\zeta} + ih(\bar{\zeta})$. Use $\bar{z} = \bar{\zeta} - i\overline{h(\zeta)} = \bar{\zeta} - ih(\bar{\zeta})$.
5. In Exercise 4 take $h(\zeta) = \zeta^2$, obtain $z^* = -\bar{z} - i + i\sqrt{1 - 4i\bar{z}}$. Binomial series converges for $|z| < 1/4$.
6. Divide by $(z - z_0)^n$, can assume $f(z_0) \neq 0$. The reflection formula then shows $f(z)$ has no zeros inside the circle. Use reflection formula again to see it is bounded on $\mathbb{C}$, hence constant.

8. (a) Show first that as z tends to a fixed boundary circle of the annulus, the image points tend to the the same boundary circle of the image annulus. Then apply the reflection principle. The formula shows that the reflected map maps the reflected annulus onto an annulus. Continue reflecting.
9. If $b_n = 0$ for n odd, then $\psi(z) = z - b_2z^2 + b_4z^4 - b_6z^6 + \cdots$ maps the imaginary axis onto γ , so ψ^{-1} is analytic at 0 and straightens out the angle.

Chapter XI: Conformal Mapping

Section XI.1

1. $w = (z^{3/2} - s)/(z^{3/2} + s)$, for any $s > 0$. Though map is not unique, the sketch is.
2. $w = -i(\sqrt{z} - 1)/(\sqrt{z} + 1)$.
3. $w = (2A/\pi i) \operatorname{Log}((1 + iz)/(1 - iz))$.
4. $w = (1 + \zeta)/(1 - \zeta)$, where $\zeta = -e^{\pi(1+i)z/2}$. Median is mapped to top half of unit circle.
5. $w = -(z + i)^2/(z - i)^2$, $w(1) = 1$.
7. $w = (-i\zeta^2 + 2\zeta - 1)/(\zeta^2 - 2\zeta - 1)$, where $\zeta = z^{\pi/2\alpha}$ brings it to a lune.
8. $\zeta = (z - b)/(1 - bz)$, $\eta = -(\sqrt{\zeta} + i)^2/(\sqrt{\zeta} - i)^2$, $\xi = i(\eta - i)/(\eta + i)$, $w = (\xi - a)/(1 - a\xi)$, where $-1 < a < 1$.
12. $w = -(\sin z - i)/(\sin z + i)$, $w(\infty) = -1$.

Section XI.2

2. $\varphi_r(z) = z/r$.
3. Convert a connected component E of $\mathbb{C}^* \setminus D$ to a point or closed analytic curve by applying the Riemann mapping theorem to $\mathbb{C}^* \setminus E$.

Section XI.3

3. Since $1 - t/a \rightarrow 1$ uniformly on bounded sets as $a \rightarrow \infty$, $g_a(z) \rightarrow \int_0^z t^{\alpha-1}(1-t)^{\beta-1}dt / \int_0^1 t^{\alpha-1}(1-t)^{\beta-1}dt$ uniformly on bounded sets. This is the Schwarz-Christoffel formula in the case that $g(\infty) = w_0$.
6. $g(z) = A \int_1^z \zeta^{-1} d\zeta$.
7. $w'(z) = c(z+1)^{-1/2}(z-1)^{-1/2} = c/\sqrt{1-z^2}$, $w = c \sin^{-1} z$, $z = \sin(w/c)$, $c = 2a/\pi$.
8. (a) $g(z) = \tau + A \int_0^z (t-a)^{\alpha-1} t(t-b)^{-a} dt$, where $\pi\alpha$ is the angle from the negative real axis to $[0, \tau]$. (c) The map in (b) is a special case of the map in (a) in which the z -plane is scaled so that the two vertices corresponding to 0 are mapped to $1 - \sigma$ and σ , where $0 < \sigma < 1$.
9. The Schwarz-Christoffel formula can be integrated, and $\pi g(z)$ is the appropriate branch of $\sqrt{z^2 - 1} + \log(z + \sqrt{z^2 - 1})$.
10. $g'(z) = A(z+1)^{-1}(z-1)^{-1}z^{-1/2}$, $g(z) = (2/\pi) \tan^{-1} \sqrt{z} + (1/\pi) \log((1 + \sqrt{z})/(1 - \sqrt{z}))$.

Section XI.4

3. $g'(z) = Az^{-1}(z-1)(z+1)$, $g(z) = (i/\pi)(z^2 - 2 \operatorname{Log} z - 1) - 1$. Stream function is $\operatorname{Arg} h(w)$.

Section XI.5

1. Use the fact that if every subsequence has a subsequence that converges to the same limit, then the sequence converges. In this case, any normally convergent subsequence has limit 0.
3. Consider the translates $f_n(z) = f(z + n)$.
4. (a) Consider the dilates $f_n(z) = f(nz)$.
6. Show first that $f_n(z) \rightarrow z_0$ uniformly on some disk centered at z_0 . See Exercise IX.1.8.
7. The n th iterate $f_n(z)$ of $f(z)$ satisfies $f'_n(z_0) = f'(z_0)^n$. Since the f_n 's are uniformly bounded on D , the derivatives of the f_n 's are uniformly bounded on each compact subset of D .

Section XI.6

1. $g(\zeta) = (\zeta - b)/(1 - \bar{b}\zeta)$, similar formula for $f(\zeta)$, and $|(f \circ h \circ g)'(0)| = (t + 1/t)/2$ for $t = \sqrt{|b|}$.
2. Apply the Schwarz lemma to $f \circ \varphi^{-1}$.
5. Compose with conformal map of D_1 onto $\mathbb{D}$, can assume $D_n \subset \mathbb{D}$ and $w_0 = 0$. Then the g_m 's are uniformly bounded. By the Hurwitz theorem, any normal limit is either constant or univalent.
7. $\rho(f(z), z_0) \approx 2|f(z) - z_0|/(1 - |z_0|^2)$ and $\rho(z, z_0) \approx 2|z - z_0|/(1 - |z_0|^2)$.

Chapter XII: Compact Families of Meromorphic Functions**Section XII.1**

6. Map $\mathbb{C}^* \setminus E$ conformally onto $\mathbb{D}$, apply this version of Montel's theorem.
8. Show $f_n(z)$ eventually has same number of zeros as $f(z)$.

Section XII.2

3. Apply Montel's theorem.
6. (b) Assume D is a disk. Consider family of square roots with appropriate branches.
7. (a) Use Exercise 6.
8. Suppose theorem fails for $f_n(z)$ and constant $1/n$. Show first that $\{f_n(z)\}$ is not a normal family, then apply the Zalcman lemma.

Chapter XIII: Approximation Theorems**Section XIII.1**

3. Exhaust D by appropriate compact sets K_n , construct $f_n(z)$ and $h_n(z)$ by induction so that $h_n(z_j) = 0$ for $1 \leq j < n$, $h_n(z_n) = 1$, and $h_n(z)$ is small on K_{n-1} . Set $f_n(z) = f_{n-1}(z) + (w_n - f_{n-1}(z_n))h_n(z)$, $f(z) = \lim f_n(z)$.

Section XIII.2

9. $(\pi/3z^2)[\cot(\pi z) + \lambda \cot(\lambda\pi z) + \lambda^2 \cot(\lambda^2\pi z)]$, where $\lambda = e^{2\pi i/3}$.

Section XIII.3

1. (a) 2, (b) $1/2$, (c) 4.
9. Use the product expansion of $\sin(\pi z)$.

Section XIII.4

3. $z \prod_1^\infty (1 - z^2/\sqrt{n}) \exp(z^2/\sqrt{n} + z^4/2n)$.

6. $f(z) = c \prod_0^\infty (1 - z/2^k)$.

11. Take $g(z)$ to be analytic with simple zeros at the z_j 's, take $h(z)$ to be meromorphic with simple poles at the z_j 's and residue $\lambda_j/g'(z_j)$, set $f(z) = g(z)h(z)$.

Chapter XIV: Some Special Functions**Section XIV.1**

6. Combine Exercises 4 and 5.

Section XIV.2

6. (b) Substitute $s = iz$.

8. Recall Cauchy-Hadamard formula for radius of convergence.

12. Cauchy estimates show that if $f(z) = \mathcal{O}(|z|^\alpha)$ in S , then $f'(z) = \mathcal{O}(|z|^{\alpha-1})$ in S_ϵ .

Section XIV.4

13. Suppose $\sigma_a = 0$, and $f(s)$ extends analytically across 0. Express power series $\sum f^{(k)}(1)(s-1)^k/k!$ as double series, substitute $s = -\delta$, justify interchanging order of summation, and sum.

Chapter XV: The Dirichlet Problem**Section XV.1**

2. Use Green's first formula.

3. Use Exercise 2.

4. Use Green's first formula, with u and v interchanged.

5. Apply Exercise 4 to u and $w = v - u$.

Section XV.2

5. (a) See the Taylor series expansion in Exercise IV.8.6.

Section XV.3

5. (c) $d(z, w)$ coincides with the hyperbolic distance $\rho(z, w)$. Use Harnack's estimate to determine $d(0, r)$, then use conformal invariance.

Section XV.4

1. Try a linear combination of 1 and $\log |z|$.

2. Use $\epsilon \log |z|$.

Section XV.5

1. Compose with Riemann maps to reduce to the case where $E_0 = \{|z| = 1\}$ and D is the domain between the unit circle and a simple closed analytic curve E_1 in $\{|z| > 1\}$. In this situation, show that $v(z)$ is strictly increasing on E_0 , and choose α so that the increase of $\alpha v(z)$ around E_0 is 2π .

Section XV.6

6. Use Green's third formula.

Section XV.7

10. Express D as an increasing union of bounded domains U_m , each bounded by a finite number of piecewise smooth closed curves, let g_m be Green's function for U_m , and define $D_m = \{g_m(z) > \varepsilon_m\}$.

Chapter XVI: Riemann Surfaces**Section XVI.1**

2. Use two annuli as coordinate patches.

References

Books

- L. Ahlfors, Complex Analysis (3rd ed.), McGraw-Hill, 1979.
L. Ahlfors, Conformal Invariants, McGraw-Hill, 1973.
L. Carleson and T. Gamelin, Complex Dynamics, Springer-Verlag, 1993.
J.B. Conway, Functions of One Complex Variable, Springer-Verlag, 1973.
J. Dieudonné, Foundations of Modern Analysis, Academic Press, 1960.
W. Kaplan, Advanced Calculus, Addison-Wesley, 1952.
K. Knopp, Theory of Functions (Parts I and II) and Problem Book (Volumes 1 and 2), Dover, 1951.
N. Levinson and R. Redheffer, Complex Variables, Holden-Day, 1970.
Z. Nehari, Conformal Mapping, McGraw-Hill, 1952.
H.-O. Peitgen and P.H. Richter, The Beauty of Fractals, Springer-Verlag, 1986.
A. Pfluger, Theorie der Riemannschen Flächen, Springer-Verlag, 1957.
R. Remmert, Funktionentheorie (Vol. I and II), Springer-Verlag, 1984 and 1991.
J.-P. Serre, A Course in Arithmetic, Springer-Verlag, 1973.
G. Springer, Introduction to Riemann Surfaces, Addison-Wesley, 1957.
E.C. Titchmarsh, The Theory of Functions (2nd ed.), Oxford University Press, 1939.
M. Tsuji, Potential Theory in Modern Function Theory, Maruzen, 1959.

Lecture Notes and Articles

- P. Chernoff, Pointwise convergence of Fourier series, Amer. Math. Monthly 87 (1980), pp. 399–400.
S. Gál, Lectures on Number Theory, Jones Letter Service, 1961.
T. Gamelin, review of *A History of Complex Dynamics, from Schröder to Fatou and Julia*, by D. Alexander, Vieweg, 1994; in *Historia Math.* 23 (1996), pp. 74–84.
D. Zagier, Newman's short proof of the prime number theorem, Amer. Math. Monthly 104 (1997), pp. 705–708.
L. Zalcman, Normal families: new perspectives, Bull. Amer. Math. Soc. 35 (1998), pp. 215–230.

List of Symbols

$\mathbb{Z}$	integers
$\mathbb{R}$	real line
$\mathbb{C}$	complex plane
$\mathbb{C}^*$	extended complex plane = $\mathbb{C} \cup \{\infty\}$, Riemann sphere
$\mathbb{H}$	upper half-plane, $\{\operatorname{Re} z > 0\}$
$\mathbb{D}$	open unit disk, $\{ z < 1\}$
$\mathcal{J}$	Julia set (Section XII.3)
$\mathcal{M}$	Mandelbrot set (Section XII.5)
$\mathcal{L}$	Laplace transform (Section XIV.2)
Re	real part of (Section I.1)
Im	imaginary part of (Section I.1)
Arg	principal branch of argument (Section I.2)
Log	principal branch of logarithm (Section I.6)
Res	residue (Section VII.1)
$\Gamma(z)$	gamma function (Section XIV.1)
$\zeta(s)$	zeta function (Section XIV.3)

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